Enhancing RAG with Domain-Specific Knowledge Graphs for Accurate Medical Data Retrieval

Saumya Dabhi, Faryaneh Poursardar
Virginia Modeling, Analysis and Simulation Center
Old Dominion University
Suffolk, Virginia, USA
(s1dabhi, fpoursar) @odu.edu

***Abstract*—This study explores the integration of a domain-specific knowledge graph (KG) into a Retrieval-Augmented Generation (RAG) pipeline to improve the retrieval of medical information. We constructed a knowledge graph from unstructured diabetes-related text. Two experimental setups were compared: a standard RAG model using raw text retrieval, and a graph-enhanced RAG model that retrieves information based on the structured graph queries. Results indicate that incorporating the knowledge graph significantly improved the retrieval process by getting more in-depth information from graph.**

***Index Terms*—Knowledge Graph (KG), Retrieval Augmented Generation (RAG), Large Language Model (LLM)**

I. Introduction

LLMs have significantly advanced AI by enabling machines to generate human-like responses across various domains. While powerful, their reliance on static training data limits their ability to access new or evolving information, often resulting in hallucinations. One way to avoid this is to fine-tune it on domain-specific tasks [1]. However, this can be computationally challenging and not the optimal solution.

To address this limitation, RAG combines LLMs with external document retrieval to enhance factual accuracy during inference. However, RAG systems often retrieve entire documents that, while partially relevant, contain large amounts of unrelated or noisy information. For example, a retrieved document may match the query on one section, but also include unrelated content that is not useful for answering the question. Since the entire document is passed to the LLM, this irrelevant information adds unnecessary processing load and can potentially dilute the quality of the generated response.

To improve retrieval, a better approach is introduced by integrating structured sources like knowledge graphs (KGs), which represent entities and their relationships in a semantically rich format. In this study, we incorporated a domain-specific knowledge graph into the RAG framework using diabetes-related data. By guiding retrieval through structured relationships, our approach improves information retrieval and reduces irrelevant information. We evaluate the system's performance with and without the knowledge graph to highlight its impact on medical information retrieval.

II. Related Work

Recent studies have demonstrated the benefits of incorporating a knowledge graph in information retrieval and generation tasks. The KG-RAG framework improves context processing and reduces hallucinations in LLMs by utilizing a Chain of Explorations (CoE) algorithm for systematic KG traversal [2]. Additionally, Graphusion [3] is a RAG framework that constructs scientific KGs, effectively capturing complex relationships to boost LLM performance. Furthermore, KRAGEN [4] enhances RAG for biomedical tasks by integrating KGs which improves the accuracy and reliability of generated responses. Optimizing prompt generation for biomedical LLMs using KGs emphasizes the importance of structured knowledge in enhancing output precision [5].

Our work builds upon this direction by integrating a domain-specific knowledge graph focused on diabetes into a RAG pipeline. We extract and connect key medical concepts and use this structured knowledge to guide retrieval alongside traditional vector search.

III. Methodology

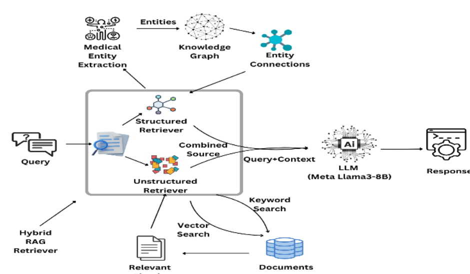

Fig. 1. End-to-end framework

This experiment uses unstructured text data related to diabetes, sourced from 7 PDF documents: 5 research papers and 2 official guideline documents from the Centers for Disease Control and Prevention (CDC). To create a knowledge graph from these documents, we used GPT-3.5 and stored this graph representation of data in a Neo4j graph database. This forms the structured component of the system, which captures relationships and connections between key concepts. Additionally, the structured component includes a lightweight module powered by GPT-3.5, which is responsible for extracting medical entities from the query. We prompt GPT-3.5

2835-5776/25/$31.00 ©2025 IEEE
DOI 10.1109/IRI66576.2025.00031

TABLE I
RESPONSE COMPARISON WITH KG TRACE

Question	With KG	KG Trace	Without KG
How does alcohol intake affect diabetes?	- Increasing the risk of hypoglycemia and/or delayed hypoglycemia, particularly for those using insulin or insulin secretagogue therapies - Weight gain - Increasing the risk of hyperglycemia, especially for those consuming excessive amounts - Interacting with Sglt2 Inhibitors, leading to excessive alcohol intake - Being advised against exceeding recommended daily limits, especially for adults with diabetes	* Alcohol Intake → Weight Gain * Alcohol Intake → Hyperglycemia * Alcohol Intake → Hypoglycemia * Alcohol Intake → Delayed Hypoglycemia * Alcohol Intake - ADVISE_AGAINST_EXCEEDING → Adults With Diabetes * Abstainers - ADVISE_AGAINST_STARTING → Alcohol Intake * Sodium–Glucose Cotransporter 2 (Sglt2) Inhibitors - AVOIDANCE → Alcohol Intake * Sglt2 Inhibitors - INTERACTION → Excessive Alcohol Intake	- Hypoglycemia and/or delayed hypoglycemia (particularly for those using insulin or insulin secretagogue therapies) - Weight gain - Hyperglycemia (for those consuming excessive amounts)
How Sugar affects diabetes?	High blood sugar can cause a range of symptoms, health problems, and complications, including heart disease, vision loss, kidney disease, and complications of diabetes. It can also damage the body over time. Insulin resistance, which is often associated with high blood sugar, can lead to type 2 diabetes if left untreated.	* High Blood Sugar → Heart Disease, Vision Loss, Kidney Disease * High Blood Sugar → Complications Of Diabetes * Insulin Resistance → Type 2 Diabetes * Blood Sugar → High Blood Sugar	High blood sugar is damaging to the body and can cause serious health problems, such as heart disease, vision loss, and kidney disease.

to specifically find diabetes related entities. These extracted entities are then used to query the KG. We utilized cypher query language to query the KG.

For the unstructured component, we extracted the text from the documents and split it into segments of 1,000 characters with a 200-character overlap to reduce information loss. This component enables the retrieval of relevant documents using keyword and vector-based search methods.

When the KG component is enabled, the query is processed through both the structured (KG) and unstructured (vector store) retrieval pipelines. This setup retrieves both semantically connected entities from the KG and relevant documents from the vector store. The combined context, along with the original query, is then passed to an LLaMA 3 model with 8 billion parameters for response generation. The output of this configuration is shown in Table 1 under the column titled ”With KG.” In contrast, when the KG component is disabled, only documents retrieved from the vector store are used as context for the LLM. The resulting output in this case is shown in the column labeled ”Without KG.”

IV. RESULT AND EVALUATION

As shown in Table I, the responses generated with the KG are more detailed and informative than those generated without it. The KG improves the retrieval process by deeply exploring the relationships between concepts linked to the query. The system begins by identifying the main topic of the query and then navigates through the graph, extracting connected nodes and edges to uncover all relevant relationships. This can be seen in the KG trace column of Table I.

In contrast, traditional RAG retrieves document chunks based solely on vector similarity between the query and the documents. This can lead to the inclusion of irrelevant information that is not directly related to the query. By analyzing the knowledge graph trace, it becomes clear that the system using the graph retrieves more focused and in-depth information, which is often missed when relying only on the standard RAG.

V. CONCLUSION AND FUTURE WORK

We integrated diabetes-focused knowledge graphs into our RAG pipeline and saw notably better information extraction by structuring unstructured data and defining clear concept relationships. Our future task is to expand the knowledge graph to include a broader array of medical domains beyond diabetes. Because raw graph traces can be hard to interpret, we will next experiment with different LLMs to build higher-quality graphs and translate node-edge data into plain language before passing it to the LLM. Finally, we’ll refine the LLM’s output clarity since our current work prioritized retrieval, so response readability remains to be optimized.

VI. ACKNOWLEDGMENT

This work was supported in part by the National Science Foundation (NSF) under Grant No. 2426781.

REFERENCES

[1] Saumya Dabhi, Joseph Martinez, and Faryaneh Poursardar. “Evaluating the Effectiveness of Fine-Tuning Large Language Model for Domain-Specific Task”. In: *2024 IEEE International Conference on Information Reuse and Integration for Data Science (IRI)*. IEEE. 2024, pp. 176–177.

[2] Diego Sanmartin. “Kg-rag: Bridging the gap between knowledge and creativity”. In: *arXiv preprint arXiv:2405.12035* (2024).

[3] Rui Yang et al. “Graphusion: a RAG framework for Knowledge Graph Construction with a global perspective”. In: *arXiv preprint arXiv:2410.17600* (2024).

[4] Nicholas Matsumoto et al. “KRAGEN: a knowledge graph-enhanced RAG framework for biomedical problem solving using large language models”. In: *Bioinformatics* 40.6 (2024), btae353.

[5] Karthik Soman et al. “Biomedical knowledge graph-optimized prompt generation for large language models”. In: *Bioinformatics* 40.9 (2024), btae560.